

Try-with-Resources

Automated Resource Management

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The Concept of Resource Closing

In Java, resources such as file streams, network sockets, or database connections must be closed once they are no longer needed. Failure to do so leads to **Resource Leaks**, where memory or file handles remain occupied.

Rule: Any object that implements the `java.lang.AutoCloseable` interface (including all `java.io.Closeable` objects) can be used with try-with-resources.

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1. The Legacy Way (Pre-Java 7)

Before Java 7, developers had to manually call `.close()`, often in a `finally` block, to ensure the resource was released even if an error occurred.

```
FileOutputStream output = null;
try {
    output = new FileOutputStream("data.txt");
    output.write("Hello".getBytes());
} catch (IOException e) {
    e.printStackTrace();
} finally {
    if (output != null) output.close(); // Messy boilerplate
}
```

2. The Modern Way (Try-with-Resources)

Java 7 introduced a cleaner syntax where the resource is declared inside parentheses after the `try` keyword. Java automatically handles the closing.

```
try (FileOutputStream output = new FileOutputStream("data.txt")) {  
    output.write("Hello".getBytes());  
    // No close() call needed!  
} catch (IOException e) {  
    System.out.println("Error writing file.");  
}
```

Multiple Resources: You can open multiple resources by separating them with a semicolon:

```
try (Scanner sc = new Scanner(file); PrintWriter pw = new PrintWriter(out))  
{ ... }
```

3. Why Use It?

- ✓ **Safe:** Guarantees closure even if a crash occurs mid-block.
- ✓ **Clean:** Removes the "Boilerplate" code associated with `finally` blocks.
- ✓ **Readable:** Keeps the focus on the actual business logic rather than housekeeping.

Efficient Resource Management with

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Less Boilerplate. More Logic.